

ASPEN WANG

 aspenwang.github.io  aspenwang  aspenwang

TECHNICAL SKILLS

Python, PyTorch, deep learning, model evaluation, $\LaTeX$

EDUCATION

Vassar College

B.A. in Computer Science; Minor in Statistics

Aug. 2022 – May 2026

Poughkeepsie, NY

RESEARCH EXPERIENCE

Research Intern

May 2025 – Present

Department of Electrical Engineering and Computer Science, University of Tennessee at Knoxville

Remote

- Advisor: Dr. Fnu Suyu
- **Focus:** Reliable machine learning, uncertainty estimation, model calibration
- Investigated how machine learning models behave under ambiguous human labels, focusing on uncertainty, calibration, and reliability beyond standard accuracy metrics.
- Analyzed failure modes of confidence-based pseudo-labeling in semi-supervised learning, showing that high model confidence does not always imply reliable supervision.
- Designed and evaluated representation-based strategies for selecting more reliable unlabeled examples under noisy or ambiguous supervision.
- Built large-scale PyTorch experimentation pipelines for training, evaluation, and statistical analysis across multiple datasets, model architectures, augmentations, and calibration metrics.
- Synthesized experimental findings into a co-authored manuscript submitted to NeurIPS 2026 for review.

Summer Research Assistant

May 2025 – Aug. 2025

Department of Chinese and Japanese Studies, Vassar College

Poughkeepsie, NY / Hybrid

- Advisor: Dr. Pier Carlo Tommasi
- **Focus:** Text mining, semantic analysis, topic modeling, computational social science
- Developed a custom NLP pipeline for premodern Japanese texts, addressing tokenization, normalization, and representation challenges in noisy historical language data.
- Designed preprocessing and embedding-based representation strategies for low-resource, high-ambiguity textual data.
- Evaluated topic modeling approaches including LDA and BERTopic, comparing traditional probabilistic methods with embedding-based semantic models.
- Applied computational analysis to study thematic structure in historical texts, connecting language modeling methods with broader questions in social and cultural analysis.

Summer Research Intern

May 2024 – Aug. 2024

Depositar Lab, Institute of Information Science, Academia Sinica

Taipei, Taiwan

- Advisor: Dr. Tyng-Ruey Chuang
- **Focus:** Data engineering, research data management, digital preservation
- Designed and implemented reusable ETL workflows for large-scale digital archives, emphasizing data integrity, reproducibility, and FAIR data principles.
- Analyzed research data management frameworks and synthesized best practices for scalable, interoperable research data systems.
- Refactored a legacy JavaScript application into a modular package, improving maintainability and supporting broader use of research infrastructure tools.
- Collaborated on infrastructure supporting real-world research workflows, bridging data engineering, metadata design, and downstream analysis.

PUBLICATIONS

Co-author, *Anonymized manuscript on semi-supervised learning under label ambiguity*. Submitted to NeurIPS 2026 for review.

TEACHING

Teaching Assistant for C MPU 145: Foundations of Computer Science	Sep. 2025 – Present
Teaching Assistant for C MPU 331: Compilers	Sep. 2025 – Present

PROFESSIONAL EXPERIENCE

Digital Initiative Intern , Vassar College Library	Sep. 2023 – May 2025
Computer & Information Service Intern , Vassar College CIS Center	Sep. 2023 – May 2025

LANGUAGES

English; Mandarin Chinese